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PROGRESSION

The PSIC Learning Approach begins with an underlying philosophy that every skiing experience should be approached as an **ADVENTURE** that is built through **Fun, Safe, Learning**... and each and every step along its progression must be thought of as a **milestone moment**, important in its own right. Each step is important technically, in terms of building skill (learning) through time-on-task practices, but also important psychologically in terms of nurturing a sense of security and confidence (safety), a level of enjoyment and the feeling of accomplishment (fun). The PSIC Learning Approach aims to provide all of the necessary tools for a rapid progression to parallel skiing delivered in a supportive and fun environment, with the central aim of converting **Day One Skiers** into **Life Long Enthusiasts**.

Early Mobility Training

Early Mobility Training (EMT) is an essential part of familiarizing students with the cumbersome nature and restrictive properties of the equipment they'll be wearing for their first ski experience. Skis will feel long and awkward at first, boots heavy and restrictive. EMT should aim to guide students through various tasks which are designed to compensate for the anatomical restriction imposed by their equipment. These tasks should be simple in nature, performed on flat terrain, in a well-defined and safe area of the beginner zone. By the end of this session, students should feel comfortable with the equipment they are wearing, understand its function and feel excited and ready to begin sliding.

TASKS MAY INCLUDE:

WALKING IN A CIRCLE ROCKING HEEL TO TOE
– with emphasis on how much the knee must work to compensate for ankle restriction.

LATERAL STANDING JUMPS
– with emphasis on co-ordination of knee and hip joint as main source of power for each jump, as well as placing the boot on edge for better grip during take-off and landing.

BOW TIE WITH BOOT
– with emphasis on how the leg is responsible for the turning effort.

JUMP STEPPING
– with emphasis on placing the landing foot accurately in front so that balance is maintained against the forces at play. For an added challenge, attempt to land on one foot without support from poles or the free foot.

TWO FOOT HOPPING
– with emphasis on coordinating knee and hip flexion to compensate for restricted ankle mobility.

ONE FOOT HOPPING
– with emphasis on maintaining pinpoint accuracy (for/aft & lateral balance) for a clean landing each time. Poles may be use for stabilization or removed for an added challenge.

Free Foot Glide (one ski on, one ski off)

The **free foot glide** offers new skiers the best of both worlds as we move them into their first gliding experience. It offers a taste of the low friction properties of the ski on snow, while providing them the security of a "free foot" which can be used to stabilize, break or push themselves in any given direction. Early skill building begins here with the **free foot glide** which can be used across multiple applications.

Some examples include:

Micro glides – alternate between stepping and gliding on flat terrain. This will be their first time feeling how the ski feels against the snow. New skiers are often surprised at how slippery this interaction is, so take time here to ensure students are comfortable with the sensation before moving on.

Long glide challenge – see how far your students can glide without touching down. This challenge will begin honing for/aft and lateral balance. Throughout this task they will become more accustomed to the unusual sensation of standing over a less stable base of support (BOS) while on the move. Poles can be used as a tool to fine tune their balance and the weight of their free foot used as a counter balance.

Free foot hockey-stop – have your students push into a glide, plant the free foot as an anchor, pivot the ski into a drift and then edge it to grip. This will be the first time that students gain a sense that the ski itself can be used as a tool to produce stability, grip and momentum control. This exercise is relatively sophisticated mechanically, but the advantages given by the free-foot are usually enough to have new skiers succeeding within a few attempts. Practice this task several times until students can produce basic grip control on each foot.

Two-Foot Glide (both skis on)

The most important consideration when the time comes to introduce a **two-foot glide** is the instructor's choice of terrain. The flatter the better! Now that we have eliminated the free foot "safety net", and without having acquired the technical means by which they can yet control speed or direction change, it's likely that the student will feel somewhat exposed during this stage of the lesson. It's for this reason that terrain choice is critically important. A flat and smooth area with a slight incline on one side is optimal.

Shuffling on the flats

- Will give them a first sense of the difference in stability

Walk around the world (stepping around a defined circle)

- Create a slight wedge with every step
- Encourage a rhythmical, consistent cadence
- Involve the poles to assist

Pole pushing on the flats

- Poles in the snow behind boots

Herringbone on the flats

- Duck-foot/knock-knee stance for edge grip

Sidestep up slight incline

- Keep skis parallel
- Across fall line
- Angulate for grip

Handbrake turn – place poles downhill and step skis to fall line

- Place poles downhill and step skis to fall line
- Keep arms rigid to hold pressure through poles
- Release pressure through arms to begin gliding

Gliding straight run with focus on:

- Even weight on both feet
- Relaxed and centered posture
    - Appropriate flex in joints
    - Arms out to the side dragging poles

⚠ Common Problems

- **Stance Collapse (falling back at onset of motion)**
- **Over reaction / compensating with stance / complete loss of balance**
- **Hunched over stance**
- **Lack of coordinating for handbrake turn**

The Braking Snowplow

The Braking Snowplow is arguably the most important task along the entire progression because it contains all the mechanical elements of more advanced skiing and will serve to instill an all important sense of control within your student. Despite the fact that they may only be controlling speed along a single trajectory, successfully achieving a braking plow is an incredibly important milestone psychologically and should be celebrated accordingly. When performed correctly, a student's ski tips should remain consistent in their proximity to one another, tails will spread apart, and the skis edge angle will increase slightly. **Time on task** is essential here to ensure the relevant skills are locked in before formally introducing direction change. Achieving good command of the skills needed for a successful **Braking Snowplow** will prove vital as we advance them towards more difficult tasks on steeper & more challenging terrain.

Explanation and stationary practice on flat terrain

- Spreading of the feet and skis in order to increase edge angle
- Twisting of the legs and skis in order to create a wedge shape and subsequent opposing steering angle
- Subtle flexing of the joints in order to avoid "locking out" will also enhance stance versatility and mobility

The snowplow push

- Pushing with poles on the flats while shaping a wedge with their skis

Explanation and first attempt on gentle slope

- Combine the movement of spreading the feet and turn legs inwards
- As skis begin to grip, increased knee flexion will allow skier to balance precisely against the "slowing" force
- **Coaching in reverse** will offer the best options for students struggling with this task and allows for visual/verbal and hands-on coaching as needed

⚠️ Common Problems

- **Skis spread with no twist resulting in wide track parallel**
- **Skis twist with no spread resulting in crossed tips**
- **Skis twist and spread with little to no edge angle (bowed legs) resulting in a lack of grip and minimal speed control**
- **Skis edge with little or no twist or spread (knock knees)**

Single Turn (Isolated) Direction Change

Now that our students have developed a state of control over their speed and momentum along a single trajectory, it's time to give them control over their direction with a **Single Turn Direction Change**. Once they are able to control their speed as well as their direction, they are well on their way to identifying as full-fledged skiers, and this is an accomplishment that we should celebrate with them and something to be very proud of!

Explanation and first attempt on gentle slope

With the asymmetrical coordination required for a **Single Turn Direction Change** things begin to get a little more intricate, so it's important at this stage to keep our explanations as simple as possible and avoid confusing students with overly complex instructions, terminology or jargon. Instead, you might inform your student on "what" to do, while leaving the "how" to do (to a certain extent) up to their intuition. For example, "ski straight towards me using your braking plow, and when you feel ready, simply turn your skis a little to the left".

- → Avoid overcomplicating the explanation
- → Have students begin with The Braking Snowplow (as practiced above)
- → Once they are in motion, simply ask them to turn their skis in one direction
- → Assess one of two possible outcomes:

Outcome One

- Skis turn in the intended direction
- Identify primary turning method - pivot, edge or pressure added to outside ski
  - Repeat the task with positive reinforcement
  - Begin to add the remaining elements as needed

Outcome Two

- Skis do not turn in the intended direction, try one of two options (if after a number of attempts your student continues to struggle, move directly to an alternative option):
  a. Offer a simple explanation/demonstration of the mechanics involved in turning outside ski (in order to increase steering angle)
     - Assess the outcome and repeat if necessary
     - Assist physically as needed, always asking first
  b. Offer a simple explanation/demonstration of the mechanics involved in "loading" pressure onto outside ski
     - Assess the outcome and repeat if necessary
     - Assist physically as needed

⚠ Common Problems

- **Outside skis turn with inadequate grip to produce direction change**
- **Outside ski locked on edge creates inability to increase steering angle**
- **Skier leans to inside of turn**
- **Skier uses excessive upper body rotation**
- **As the skier transfers weight to outside ski they also flatten the ski, so what is gained by one action is lost by the other**

Corrective solutions may include:

- Coaching in reverse
  - Ski backwards in front of student prompting instructions matched with clear demonstration
    - Offer physical support with poles if needed
  - Hands on outside knee
    - To establish better balance through center of outside ski
    - To encourage better grip through knee angulation
  - Both hands on knees
    - To promote athletic stance
    - To reduce upper body rotation
  - Tap inside ski
    - To encourage stronger balance through center of outside ski

Note that physical interactions should be avoided whenever possible. If required, always try to hold on to poles or hands only, and <u>ALWAYS ask your student before any type of physical contact.</u>

Two Turn Sequence

A **Two Turn Sequence** is a significant development milestone for our Day One Skiers as it indicates an ability to not only produce a single change in direction, but to control direction at will. Releasing out of one turn and into another will also lead to a momentary sense of exhilaration, as they flatten their edge and embrace a subtle increase of momentum. Completing a second turn will appeal to a need for control which in-turn bolsters confidence. Above all else, it's our job as instructors to successfully manage the relationship between exhilaration and control in order to optimize the skier experience, and the aim of the **Two Turn Sequence** is to bring these elements to the forefront of the **skier experience**.

Demonstration, explanation and first attempt on a gentle but slightly longer stretch of terrain.
Students may be reminded of the technique used during their single turn direction change and encouraged to let their skis run in order to increase momentum and free-up internal mechanics. Inadequate speed during this task will often lead to a lack of flow from the skis against the snow and stifle their success rate. For more timid students, encouragement may be needed to lower inhibition in order to relinquish a small measure of control to the force of gravity. Prime them to view "letting go" as an enjoyable part of their **Day One Experience**.

- Have students begin with a single turn direction change.
  - Verbal reinforcement and demonstration of the student's preferred **primary method of direction change.**
- At the completion of their first turn, encourage students to relax the tension in their legs, gently turn their skis in the opposite direction, and/or shift their balance point onto the opposing foot.
- Utilize coaching in reverse, if possible, as a way for them to mirror demonstrative actions whilst in motion. This proves particularly effective for those who struggle with bilateral coordination – marching movement patterns from one side of their body to another (left brain/right brain).

⚠ Common Problems

- **Outside ski turns with inadequate grip to produce direction change**
  - **Ski "waves" against snow, due to low edge angle or legs bowing out**
  - **Lack of rotational control and low grip pressure**
  - **Ski tips prone to crossing one another due to low edge angle leading to excessive rotational action**

- **Skis locked on edge**
  - **Outside ski locked while inside ski drifts**
  - **Both skis locked on edge (knock kneed or wide stance)**
- **Stance overly hunched with hands clenched at torso**
- **Instinctive drive leads student to turn with excessive upper body rotation toward the intended direction of travel**
- **Lacks necessary bilateral coordination to "mirror" (match) mechanics from one turn to another**

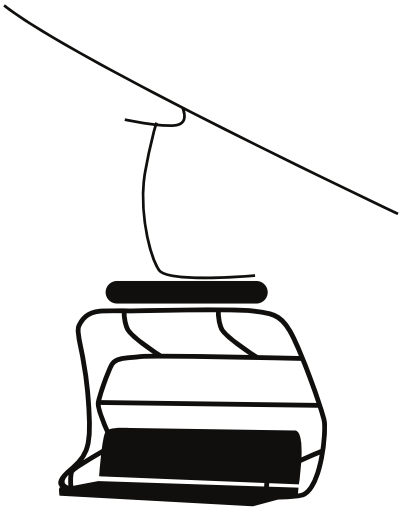
Corrective solutions may include:

- Coaching in reverse
- Aeroplane turns
- Obstacle course challenge
- Hand(s) on various outside body parts
  - Knee
  - Hip

The First Lift Experience

Up until now, for every run that your students have slid down the mountain (and for them it will feel like a mountain), they have powered themselves to the top. It's imperative at this stage of the progression to monitor fatigue levels closely and pace the remainder of the lesson according to the physical capabilities of each student. One of the biggest risks to our mission is to have students "tap-out" of their day one experience due to exhaustion. To that end, at this point of the lesson we should consider introducing our students to their **First Lifted Experience**.

The timing of this introduction will depend to a large extent on the nature of the learning area in question with respect to pitch gradient, as well as the type of lifts your resort has on hand. Magic carpets are the most common access point in the modern era, and almost all resorts have adopted them as standard. They are user friendly and significantly less intimidating than a chair lift ride, and the most obvious option for introducing our students to their **First Lifted Experience**.



Guide students to the lift line entrance or use either of the two climbing techniques introduced.
Follow your resorts guidelines for lift loading and unloading.

Multi Turn Sequence

In order to access suitable terrain for a **Multi Turn Sequence**, you'll have guided your students through their **first lift experience**, or hiked them to an elevation point high enough to allow for a number of consecutive turns. Either way, the **Multi Turn Sequence** is an opportunity to introduce students to one of the most enjoyable elements of the skier experience – skiing with rhythm. It's at this stage of the lesson that students first gain insight into how to ski with cadence in the form of rhythmically controlled turns. This adds a dimension to their experience that changes it from something that can feel somewhat mechanical to something far more free flowing and natural.

> **The multi turn sequence should begin with an explanation of the task with emphasis on linking turns directly from one to another.**
> Having students ski behind you, is an effective way to lay a pathway for them to follow. It will also give them an opportunity to match your movements from behind while keeping them in ear shot for verbal coaching and guidance. The **Multi Turn Sequence** marks a turning point to a lesson which is now "**on the move**" and from this point forward, you and your students will be skiing around the resort together.

- Have students follow your tracks
- Emphasize relevant movement patterns within your demonstration
- Offer verbal coaching "on the fly" as needed
- Use markers, cones or ski poles as visual cues around which your students can turn, in order to encourage a defined cadence
- Choose relevant drills for developing more refined steering skills within a wedge stance

Wedge Turn Versatility

As our students become confident linking rhythmical wedge turns on more expansive sections of the beginner area, the skier experience will be transformed into one of mileage, exploration and adventure. **Terrain play** becomes an essential part of raising the engagement level for those new to the sport. It is for this reason that many resorts have adopted a **terrain-based learning** approach, and at this stage, we should take advantage of any purpose built or natural terrain features that will assist in building **Wedge Turn Versatility** while enhancing the sense of adventure experienced by our **Day One Skiers**.

BANKED TURNS

– are an effective way to introduce a subtle form of inclination into their skiing. The terrain provides natural assistance by lifting their outside ski which in turn tips their **center of mass (COM)** inwards. The bank also allows their outside ski to remain relatively flat to the snow's surface which grants the skier full access to the rotary plane making the skis extremely easy to turn. The sensation of being guided around the turn by the terrain itself will be sure to provide students with an extremely fun and exhilarating ride.

ROLLERS

– are an incredibly fun and effective way to introduce students to terrain characterized by dramatic pitch variation, as it trains them to activate the major joints in order to maintain balance as terrain changes under their feet. Skiing rollers is the best way to highlight the novel sensation that rapid changes in pressure can bring to The Skier Experience – in particular, the enjoyment which arises from the feeling of weightlessness. Adults and children alike will be drawn to this sensation and utilizing rollers as a means to enhance skill, adventure and fun should be used at the earliest opportunity.

GENTLE HALFPIPES

– are a fantastic tool for promoting symmetry, timing and rhythm, all the while getting your students accustomed to the concept of a **changing fall-line** (a common terrain feature that when used to their advantage will offer significant steering assist.) The halfpipe will also amplify the effects of acceleration and deceleration, which plays perfectly against our mission of promoting the all-important balance between exhilaration and control.

RIDGELINE SKIING

– if a halfpipe provides assistance in turning skis, then a ridgeline will assist in edging them. As the terrain falls away either side of the ridge, the angle of the ski to the snow will increase by precisely the same amount. This will amplify the amount of grip generated between the ski and the snow, which trains students to manage grip without requiring excessive angulation. Ridgeline skiing will also introduce students to the idea of terrain that raises up between turns which could be considered an early intro to bump skiing.

CONE SLALOM

– will establish tangible goals against which your students can measure the success of their run. Cone configuration can be altered depending on the desired outcomes and when combined with verbal instruction and/or drills, **cone slalom** will serve as a teaching aid that is both **effective and fun**.

The First Match

There are many scenarios that can lead to a student matching their skis for the very first time, but whichever way it happens **The First Match** represents a significant **milestone moment** in their journey to parallel skiing. Up until this point our students will have relied on a wider snowplow stance, grip from the inside edge of each ski, and two opposing steering angles. Their stability thus far has come from having one foot positioned on either side of their centerline, and this is something they will often maintain into their first match. Given the appropriate speed, turn shape and terrain, students new to parallel skiing will often switch the edge of their inside ski without crossing it over the centerline. This is a perfect way for a student to bring their stance into parallel without having to rely on turning forces to hold them up.

Shifting their inside ski across the centerline amounts to a psychological and technical leap which will, from that point forward, completely transform **The Skier Experience**. However, when both skis occupy one side of the centerline, the “self centering” properties of the snowplow stance are no longer in play and a skier must now rely on external forces (centrifugal) in order to maintain balance. This can feel a little disconcerting at first, but when measured against the additional grip control that comes from having both skis on the same side of the turn, students are often quick to embrace this new sensation and seldom look back. Here are a few strategies that can get your students "across the line".

Before the inside ski can approach or cross the centerline, at some point in the turn the student must first be able to establish a balance point somewhere close to the **center of their outside foot**. Once balance has been established through the outside ski, intuition will often kick in and compel the student to drift the inside ski across to match its outside counterpart. This action can be encouraged through various means outlined here:

- **Changing external force** – a smaller wedge at higher speed.
  Given that centrifugal force pulls objects to the outside of their arc, it can be extremely useful for encouraging balance toward the outside of a ski turn. The formula that governs this force states that velocity contributes to the effect in an exponential manner, so a slight increase in speed along the arc will translate to a large shift in balance towards their outside ski. When this is combined with a smaller wedge, it can make the act (action) of shifting the inside ski towards or across the centerline almost involuntary.



↳ **Terrain choice is important for this task** – and your highest chance of success lays in a terrain profile holding a slightly steeper pitch which leads into a long flat run-out. Carefully guide your students down the steeper portion of the run. When you feel they are ready, encourage them to ease off the wedge, accelerate down the pitch and onto the flat. A **shallow turn shape** with a **narrow wedge** at **higher speed** should bring them close to the first match. For those not quite over the line, add some coaching on the fly and a strong demo in order to seal the deal.

→ **Using internal force to change balance point** – in situations where skiers rely heavily on their inside ski for stability, foot to foot balance can be altered through a number of distinct moves that new skiers can be encouraged to make with their bodies. While excessive upper body movement should never be promoted on the whole, certain movement patterns can be thought of as a means to an end if they accelerate their progression to parallel skiing. Here are a couple of options that address this issue from very different angles while achieving similar results:

↳ **Tilting upper body toward outside ski** is a move that many students find to be counter intuitive. In cases where students intuitively lean or rotate to the inside of the turn, **time on task** will be required in order for this early habit to be broken, as it will be almost impossible for a student to begin matching their skis until it is. Many different strategies can be used to change this mindset:

Strategy One

- **A simple description** of how "tilting the spine" can encourage a subtle shift of balance to the outside ski. Priming them with a "feeling" and "expected outcome" will help to connect a **technical tip** such as this to their forthcoming **real-world experience**. For the home run, add a strong demonstration accompanied with real-time verbal descriptions and your students will be armed and ready for their **first match**.

Strategy Two

- **Reference points** can be used as an effective way to encourage particular patterns of movement. "Nose over outside knee" for instance, is a great way to promote balance to the outside without the need to adopt any technical language at all. Using reference points in this way also serves to give students a more tangible method against which they can measure the success of their efforts in breaking their intuitive habit.

Inside ski lift: If asked to stand with even weight on both feet and then to lift one foot completely off the ground, a very conscious effort will be given to the action of lifting that foot off the floor. But for the foot to be lifted, a number of subconscious actions were instinctively performed beforehand – namely a subtle shift of balance and weight on to the opposing foot. Once sufficient balance has been established by the subject, the conscious effort of lifting that foot from the floor is free to kick in. This is an example of a conscious action with a subconscious "by-product", in this case one which can be leveraged to accomplish an important developmental step.

- Encourage your students to ski with a smaller wedge to minimize the lateral stance adjustment (subconscious action) needed to lift the inside ski off the surface of the snow. Once this has been established and towards the end of their turn, prompt them to begin tapping the inside ski upon the surface of the snow. If successful, challenge them to hold their ski off the snow for longer periods of time placing it closer to its outside counterpart with every tap. Continue to work in this manner until a match is made.

Part-time Parallel

Now that their center of mass has had its first taste of freedom from the constraints of its base of support, your students will feel a tangible sense of exhilaration as they enter the **Part-Time Parallel** stage of development. The training wheels are coming off, and the full potential of the vehicle that has carried them this far is beginning to come to light. From here, your goal is to develop balance independence within their skiing, where their COM and BOS are free to roam along separate trajectories – as will be required for **parallel skiing**.

However, there is a significant difference of ability between students able to match their skis for a moment at the end of a particular turn (when their COM is safely up hill of there feet) and those comfortable committing to a **high side crossover** above the fall line (where they will experience significantly more exposure). Full independence will have been achieved when the **match point** graduates from the end of the turn all the way up the to start of the turn, and from this point forward a fun and focused effort should be centered on **tasks, terrain, feedback and mileage** chosen specifically to help students match their skis higher and higher in the arc.



In order to graduate their match point towards or above the fall line, technically your students will need to develop a strong and early connection to their outside ski and be capable of producing more refined and deliberate grip control. Psychologically, they must prepare to take a "leap of faith" that the turning force they produce will hold fast against gravity in supporting their mass throughout an entire turn. This is something far easier to achieve toward the end of each turn, as the pitch inherently adds to their edge angle while naturally providing more grip. The opposite can be said for the beginning of the turn which is why students often resort back to a top turn wedge when skiing steeper, more challenging terrain. Here are a few strategies that can be used to help graduate their match point up to and beyond the fall line.

- **Convex terrain** steepens as it descends and can be extremely useful as a means of encouraging students to match their skis earlier in the turn, as it helps to **unweight, twist and edge** the skis all in consecutive order. Each one of these elements are essential in their own right for an early match – and worthy of individual focus – but when brought together in a single turn over convex terrain, positive results are almost always met.
  - **Unweighting:** As the terrain falls away from a skier's center of mass, they'll begin to feel noticeably lighter as pressure decreases under foot. This can diminish any sense of reliance felt toward their inside ski, freeing it up to drift over and match its outside counterpart.
    - Have your students follow closely in your tracks. As you lead them down, actively seek out convex terrain features while placing your turn transition on the apex of the hump. For a larger unweighting effect, you might encourage them to add active extension to the beginning of their turn.
  - **Pivoting: Convex terrain** is optimal for enabling students to initiate their turn by providing an easy surface on which to pivot their skis. As the skis roll over the crest, their point of contact will condense in towards their ski boots making the skis far easier to turn. This will lead to the production of a larger steering angle and increase their potential to create a turning force. Psychologically, this will feel very reassuring to the student and the higher in the arc that it occurs, the greater the chance they will have in matching their skis.
  - **Edging:** As the terrain falls away from the base of their ski, it will naturally increase the angle of the ski to the snow, and this, when combined with the steering angle that was just established, will provide ample turning force to hold them up against the pull of gravity. This is where the "leap of faith" will need to kick in, and to the extent that you are able, the time is now to offer moral support and strenuous encouragement!

Full Parallel Skiing

Full Parallel Skiing represents an enormous accomplishment for those new to the sport of skiing as well as the instructor(s) who have guided them through to this very impressive milestone! They have remained engaged enough through each and every step of our progression to develop the motor patters required to balance against the many forces at play while maintaining a parallel stance from turn to turn. From this point forward there is no turning back, they can proudly refer to themselves as fully fledged skiers, and the journey they have been on in getting this far, marks the beginning of a lifetime of **adventure and fun** on the slopes.

Parallel Short Turns

A well-developed parallel short turn is the gateway to an **off-piste world of adventure**. Once honed, a competent short turn will enable a student to begin navigating bumps, trees, steeps, chutes, powder and bring them to a place technically, where large-scale exploration of the greater alpine environment becomes a very real possibility.

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